

Condition Number Based Low-Bit Quantization for Image Super-Resolution

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Abstract

Low-bit model quantization for image super-resolution (SR) is a longstanding task that is renowned for its surprising compression and acceleration ability. However, accuracy degradation is inevitable when compressing the full-precision (FP) model to ultra-low bit widths ($2 \sim 4$ bits). Experimentally, we observe that the degradation of quantization is mainly attributed to the quantization of activation instead of model weights. Considering that the activation quantization error is hard to minimize, minimizing the impact of the error emerges as a good choice, which is described by the condition number. Therefore, we propose **CondiQuant**, a condition number-based low-bit post-training quantization (PTQ) for image super-resolution. Specifically, we formulate the quantization error of activation as the condition number of weight metrics. By decoupling the representation ability and the quantization sensitivity, we design an efficient proximal gradient descent algorithm to iteratively minimize the condition number and maintain the output. With comprehensive experiments, we demonstrate that CondiQuant outperforms existing PTQ methods in accuracy without computation overhead and gains the theoretically optimal compression ratio in model parameters. Our code is available at <https://github.com/Kai-Liu001/CondiQuant>.

1. Introduction

Image super-resolution (SR) aims to restore the high-resolution (HR) images from the low-resolution counterparts. It is a foundational computer vision task in low-level vision and image processing, widely studied in medical imaging (Greenspan, 2008; Isaac & Kulkarni, 2015; Huang et al., 2017), surveillance (Zhang et al., 2010; Rasti et al., 2016), remote sensing (Bandara & Patel, 2022), and mobile phone photography (Wu et al., 2024). Nonetheless, the

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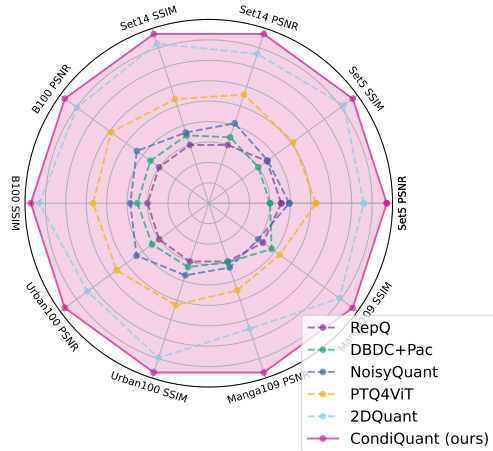


Figure 1. Comparison with SOTA PTQ methods on five benchmarks. Our CondiQuant gains consistently better performance.

existing edge devices’ notorious limited computation and memory ability hinder real-world deployment. Therefore, it is increasingly urgent to develop model compression and acceleration techniques for SR models to reduce the redundancy in both model parameters and inference computation.

Model quantization (Choukroun et al., 2019; Ding et al., 2022; Hubara et al., 2021; Li et al., 2021) is a powerful compression technique that compresses the model from full-precision to low-bit representations. With quantization, the time-consuming floating-point operations are converted into efficient integer ones, making it an ideal candidate for model compression in resource-constrained edge devices. However, the conversion inevitably leads to severe performance degradation, especially when compressing to ultra-low bit width ($2 \sim 4$ bits). The situation is more severe in vision transformers (ViTs) due to the deterioration of self-attention.

The two branches of quantization, aka quantization-aware training (QAT) and post-training quantization (PTQ), deal with the degradation in different ways. QAT, a firm adherent of backpropagation, concurrently optimizes the model weights and quantizers’ parameters via straight-through estimator (STE) (Courbariaux et al., 2016). Though inspiring results are obtained, the ineluctable demand for large datasets and long-time training discourages the deployment. Whereas PTQ is much more efficient. Most PTQ methods utilize efficient algorithms to calculate the quantizers’ parameters in a small calibration set. Given the complexity of both branches, PTQ is usually inferior to QAT and calls for

advanced solutions. To improve the inferior performance of PTQ, we observe two key experimental phenomena. The first observation is about the attribution of degradation.

Observation 1: *The degradation of quantization is mainly attributed to **activations**.*

Therefore, we leverage condition number, a metric in numerical analysis, to reflect the changes caused by the quantization of activations. In detail, condition number could measure how much the output value can change given a disturbance in the input value, *i.e.*, the quantization error of activation. Considering a small disturbance, the smaller the condition number is, the smaller the output varies. Thereafter, one way to reduce the quantization error is by minimizing the condition number of the weight matrix. Inspired by MagR (Zhang et al., 2024), we verify that their observation in LLMs also holds in SR models, which is

Observation 2: *The feature matrix X across all linear layers is approximately **rank deficient**.*

In a linear layer, the output Y can be expressed as matrix multiplication $Y = XW$, where W is the weight matrix and X is the feature matrix. When X is rank-deficient, there are infinite solutions for W that satisfy the equation. This desirable property allows a shift in the weight matrix to quantization friendliness while maintaining the output.

Based on the previous two observations, we focus on the impact of quantization of activation, establish its relationship with condition number, formulate the optimization problem, and derive proximal gradient descent to solve it efficiently. With the proposed CondiQuant, the restoration accuracy is improved evidently as shown in Fig. 1.

To sum up, our main contributions are fourfold:

1. We design CondiQuant, a novel PTQ method for image SR based on condition number. CondiQuant gains theoretically optimal compression and speedup ratios with the currently minimum performance degradation.
2. We propose that the model’s sensitivity to quantization is related to the condition number of weights in the linear layers and could be optimized efficiently.
3. We design an efficient algorithm based on proximal gradient descent to reduce the condition number of the weights while approximately maintaining the output.
4. Comprehensive comparison experiments are conducted to show the SOTA performance on efficiency and effectiveness of our proposed CondiQuant. Besides, extensive ablation studies are conducted to prove the robustness and efficacy of our proposed CondiQuant.

2. Related Work

Image Super-Resolution. The trailblazing research on image super-resolution with deep neural network is SR-CNN (Dong et al., 2016), which utilizes convolution layers to replace conventional methods in SR. Since then, dazzling network architectures (Chen et al., 2022; 2023; Lim et al., 2017; Zhang et al., 2018a) are designed to achieve more stirring image restoration accuracy. RDN (Zhang et al., 2018b) designs dense skip connection and global residual to utilize abundant local and global features. With the advance of vision transformer, SwinIR (Liang et al., 2021) uses window-attention to fully make use of global information. As reconstruction accuracy increases, researchers focus more on model parameters and operations. Generally, the ViT-based SR networks are much smaller and faster than pure CNNs. However, even the advanced SR networks like SwinIR, current SR models are still too large to be deployed on resource-constrained edge devices. Therefore, research of quantization on ViT-based networks is in urgent need to make SR possible on mobile devices.

Post-Training Quantization. PTQ is famous for its fast speed and low cost during quantization. Recently, excellent PTQ methods have been proposed to advance the performance and efficiency of PTQ. As a PTQ method specifically for ViT, PTQ4ViT (Yuan et al., 2022) proposed twin uniform quantization to reduce quantization error on the output of softmax and GELU. RepQ (Li et al., 2023) decouples the quantization and inference processes and ensures both accurate quantization and efficient inference. NoisyQuant (Liu et al., 2023) surprisingly finds that adding a fixed Uniform noisy bias to the values being quantized can significantly reduce the quantization error. However, most of the above PTQ methods are only for Transformer blocks and the generalization ability on SR tasks is relatively low. Whereas, 2DQuant (Liu et al., 2024) searches the clipping bounds in different ways on different distributions and is currently the best PTQ method on the SR task. However, most of the above methods only concentrate on the quantization process and ignore adjustments in the weights.

Condition Number. The concept of the condition number originated and developed in numerical analysis (Turing, 1948; von Neumann & Goldstine, 1947). It measures how much the output value of the system or function can change for a small change in the input. A matrix with a high condition number is said to be ill-conditioned and leads to huge output changes given a small disturbance in input. It also plays an important role in machine learning. Freund proposed optimization algorithms generally converge faster when the condition number is low (Freund et al., 2018). Additionally, the condition number is leveraged to enhance the numerical stability of regression models by fine-tuning hidden layer parameters (Xiao et al., 2018). In this work, we formulate the optimization of quantization error via condition number and propose CondiQuant to solve it efficiently.

3. Method

3.1. Preliminaries

Given an SR network, we use upper and lower clipping bounds to quantize weights and activations (Liu et al., 2024). Usually, we leverage the fake quantization to simulate the quantization process with no simulation error. The fake-quantize process can be written as:

$$x_c = \text{Clip}(x, l, u), x_r = R(s(x_c - l)), x_q = sx_r + l, \quad (1)$$

where x denotes weight w or activation a being quantized and $s = (2^N - 1)/(u - l)$ is the quantization ratio. l and u denote the lower and upper bounds for clipping, respectively. $\text{Clip}(x, l, u) = \min(\max(x, l), u)$ constrains the input to be between l and u , and $R(\cdot)$ rounds the input to nearest integer. With Eq (1), x is dispersed into discrete candidates.

3.2. Analysis

To analyze the loss attribution in quantization, we rewrite the linear layer with quantization as follows:

$$\begin{aligned} \hat{Y} &:= \hat{X}\hat{W} = (X + \delta X)(W + \delta W) \\ &= XW + X\delta W + \delta XW + \delta X\delta W, \end{aligned} \quad (2)$$

where $\hat{X}$ denotes the quantized value and δX denotes the quantization error of X . The second-order term $\delta X\delta W$ can be ignored due to its tiny impact on performance. Thus Eq. 2 can be rewritten as:

$$\hat{Y} \approx XW + X\delta W + \delta XW = Y + X\delta W + \delta XW. \quad (3)$$

Shown in Tab. 1, we observe that the influence of $X\delta W$ and $\delta X\delta W$ is smaller than δXW .

This result can be explained by two reasons. **First**, the activation varies with different inputs while the weight is fixed. Therefore, the average value of the absolute value of the δX elements is much greater than that of δW . **Second**, the W is sensitive to minor changes, *i.e.*, δX . Hence, we only concentrate on δXW and ignore $X\delta W$ in the following discussion. So we can further approximate the loss:

$$\hat{Y} \approx XW + \delta XW, \quad \delta Y \approx \delta XW. \quad (4)$$

4 indicates that the degradation of model performance can be divided into two components: (1) the magnitude of activation quantization error and the sensitivity of the weight to δ . To minimize the approximate loss, a naive solution is to reduce the Frobenius norm of δX *i.e.*, $\|\delta X\|_F$. However, it is impractical as δX is input-related, and rounding-off errors are impossible to reduce, especially with low bits. Therefore, we focus on the model’s sensitivity to quantization, related to condition number.

3.3. CondiQuant

We begin with the observation of rank deficiency of the feature matrix. Then we derive the optimization objective and illustrate the methodology to arrive at the solution.

Rank Deficiency. We visualize the rank ratio in Eq. 2. Most of the layers are rank-deficient, and only two layers are full rank. The mean value of the rank ratio is 0.626. In the linear

	Y	$Y + X\delta W$	$Y + \delta XW$	$Y + \delta X\delta W$
PSNR	32.2543	32.0117	31.3304	32.2519
SSIM	0.9293	0.9270	0.9221	0.9293

Table 1. Attribution of quantization loss. Y denotes the FP model and the rest denotes performing quantization on W , X , and both.

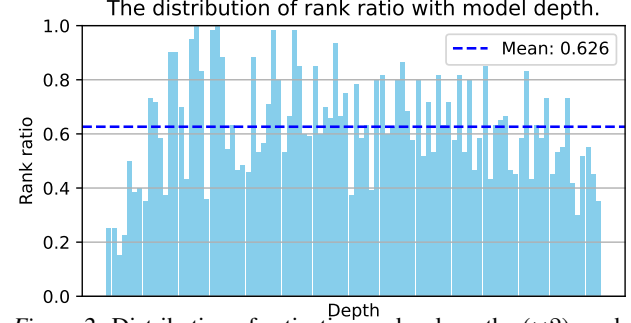


Figure 2. Distribution of activation ranks along the ($\times 2$) model depth. Most activations are severely rank-deficient.

layer, the process is matrix multiplication, *i.e.*, $Y = XW$. If W is rank deficient, there exist infinite W' that satisfies $XW = XW'$. Hence, this result serves as a theoretical foundation to select W' to minimize δXW .

Formulation. In this section, we formulate the relationship between model sensitivity and condition number. We begin with $\|\delta Y\|_2$ and $\|Y\|_2$:

$$\|\delta Y\|_2 = \|\delta XW\|_2 \leq \|\delta X\|_2 \|W\|_2, \quad (5)$$

$$\|Y\|_2 = \|XW\|_2 \geq \|X\|_2 \sigma_{\min}(W), \quad (6)$$

where $\|\cdot\|_2$ is the bi-norm of the matrix, and $\sigma_{\min}(\cdot)$ denotes the minimum singular value. With Eq. 5, we can establish the relationship between quantization loss and condition number:

$$\frac{\|\delta Y\|_2}{\|Y\|_2} \leq \frac{\|\delta X\|_2 \|W\|_2}{\|X\|_2 \sigma_{\min}(W)} = \kappa(W) \frac{\|\delta X\|_2}{\|X\|_2}, \quad (7)$$

where $\kappa(W)$ denotes the condition number of W . This formula shows that under the same rounding-off error, the impact of quantization is greater with larger $\kappa(W)$. Detailed derivation can be found in the supplementary materials. Therefore, we minimize the condition number of W and maintain the output. The objective can be written as:

$$\min_W \kappa(W) = \frac{\sigma_{\max}(W)}{\sigma_{\min}(W)}, \text{ s.t. } \|XW - X\hat{W}\|_F \leq \epsilon, \quad (8)$$

where ϵ is a small positive number that limits the magnitude of the loss. However, it is a non-convex and non-smooth optimization when minimizing the condition number directly. To address this, we construct a proxy objective to optimize and propose the proximal gradient method to solve it.

Proxy objective function. To minimize the condition number, our strategy is to compact the distribution of all singular values more densely. Specifically, we design a regularization term to minimize the deviation of singular values:

$$\min_W \frac{1}{2} \|XW - Y\|_F^2 + \lambda \cdot \mathcal{G}(W), \quad (9)$$

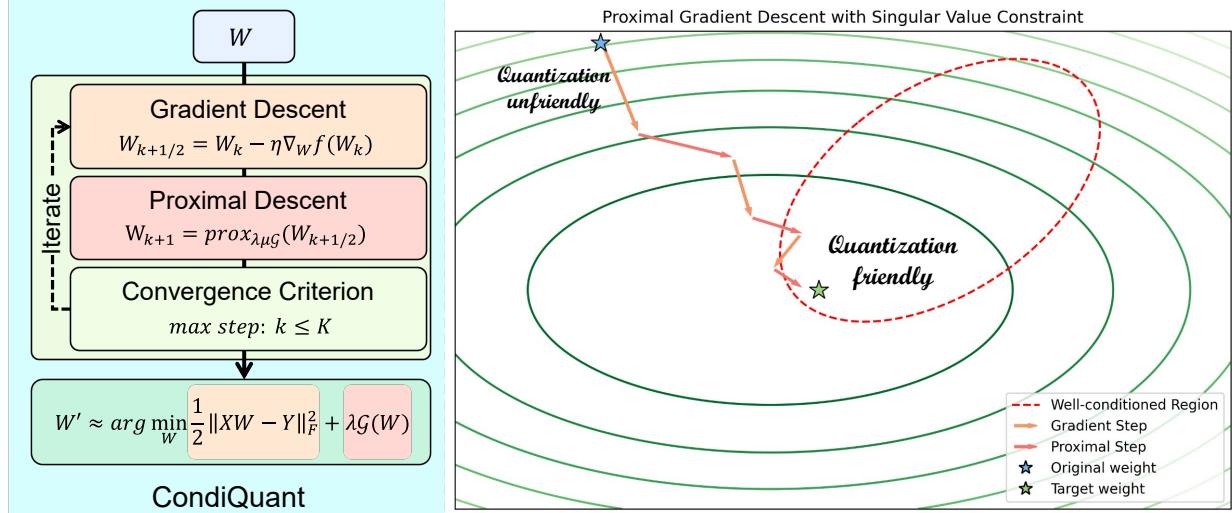


Figure 3. Overview of our proposed CondiQuant, which employs iterative optimization to minimize the condition number of the weight matrix while maintaining the output. The gradient descent step updates the weight matrix with gradients to ensure the output is close to the original. The proximal descent step minimizes the condition number with the proximal operator. The above two steps are conducted iteratively before reaching the convergence criterion, *i.e.*, the max iteration step. We illustrate the effect of both steps. With CondiQuant, the weight matrix is converted into a quantization-friendly and well-conditioned one, while the output is close to the original.

where $\mathcal{G}(\cdot)$ is a regularization term used to compact the singular value distribution and λ is the regularization strength parameter. There are several forms of $\mathcal{G}(W)$ to compact the singular values' distribution. Considering efficiency and effect, we choose the following form:

$$\mathcal{G}(W) := \sum_{i=1}^r (\sigma_i(W) - t)^2, \quad (10)$$

where $r = \text{rank}(W)$ represents the number of singular values and t is a value between $\sigma_{\min}(W)$ and $\sigma_{\max}(W)$. The median or the mean of the singular values are possible candidates for t and we choose the mean value form in CondiQuant. With this form, we could provide a closed solution to minimize $\mathcal{G}(W)$ in the following section.

Proximal Operator for $\mathcal{G}(W)$. We utilize the proximal operator to minimize the $\mathcal{G}(W)$ part. We define the proximal operator of function $\mathcal{G}(W)$ as follows:

$$\text{prox}_{\lambda \mu \mathcal{G}}(W) := \arg \min_Z \left\{ \frac{1}{2} \|Z - W\|_F^2 + \lambda \mu \mathcal{G}(Z) \right\}, \quad (11)$$

where μ is a hyper-parameter to balance the optimization. The target of Eq. 11 is to minimize $\mathcal{G}(Z)$ and keep Z close to W . To solve this problem, we perform singular value decomposition (SVD) on W to extract the singular values:

$$W = U \Sigma_W V^T, \quad (12)$$

where U and V are orthogonal matrices and $\Sigma_W = \text{diag}(\sigma_i(W))$ is the diagonal matrix of singular values. The problem could be reduced to optimize each singular value $\sigma_i(W)$ independently and can be written as:

$$\sigma_i^*(W) := \arg \min_{\sigma_i^*} \left\{ \frac{1}{2} (\sigma_i^* - \sigma_i(W))^2 + \lambda \mu (\sigma_i^* - t)^2 \right\}. \quad (13)$$

We notice that Eq. 13 is a quadratic minimization problem,

and a closed solution could be derived. Specifically, after taking the derivative and setting it to zero, we have:

$$\sigma_i^*(W) = \frac{\sigma_i(W) + 2\lambda\mu t}{1 + 2\lambda\mu}. \quad (14)$$

Using the updated singular values $\sigma_i^*(W)$, we reconstruct the matrix with lower condition number:

$$W^* = U \Sigma_{W^*} V^T, \quad \Sigma_{W^*} = \text{diag}(\sigma_i^*(W)). \quad (15)$$

To conclude, with the proximal operator, we could update W to W^* to minimize $\mathcal{G}(W)$. Hereby, the condition number of W is reduced and the output is kept still.

Proximal Gradient Descent. As discussed previously, the optimization problem contains both smooth and non-smooth components. More specifically, $\frac{1}{2} \|XW - Y\|_F^2$ is a convex and differentiable function, while $\mathcal{G}(W)$ is a convex but non-differentiable function. With the proximal operator, we design the following proximal gradient descent method to solve two components iteratively:

Step1. Gradient Descent. Update the smooth part $\frac{1}{2} \|XW - Y\|_F^2$ with the gradient descent:

$$W_{k+\frac{1}{2}} = W_k - \eta \nabla_W \frac{1}{2} \|XW - Y\|_F^2 \Big|_{W=W_k}, \quad (16)$$

where $\eta > 0$ is the step size, and ∇_W is the gradient of $\frac{1}{2} \|XW - Y\|_F^2$ at W_k . With this step, we ensure the output is as consistent as possible with the original value.

Step2. Proximal Step. Apply the proximal operator to the non-smooth part $\mathcal{G}(W)$ with Eq. 12, 14 and 15 to minimize the condition number:

$$W_k = \text{prox}_{\lambda \mu \mathcal{G}}(W_{k+\frac{1}{2}}) = U \Sigma_{W^*} V^T.$$

Step3. Repeat. Iterate the above two steps until reaching the convergence criterion, *i.e.*, $k \geq K$. k is the iteration step

(a) Ablation study on learning rate η .

η	10^{-1}	10^{-2}	10^{-3}	10^{-4}
PSNR $\uparrow$	37.13	37.15	37.15	37.08
SSIM $\uparrow$	0.9568	0.9568	0.9567	0.9564

 (b) Ablation study on target value t .

t	Mean	Median	$(\sigma_{max} + \sigma_{min})/2$
PSNR $\uparrow$	37.15	37.12	37.11
SSIM $\uparrow$	0.9567	0.9566	0.9567

 (c) Ablation study on λ .

λ	0.001	0.002	0.004	0.005	0.010
PSNR $\uparrow$	37.13	37.12	37.13	37.10	37.08
SSIM $\uparrow$	0.9567	0.9566	0.9567	0.9566	0.9565

Table 2. Ablation studies of CondiQuant on Set5 ($\times 2$, 2 bits). The selection of hyperparameters and the effect of CondiQuant are evaluated. The results demonstrate that CondiQuant is robust and improves performance.

while K is the total step of iteration. We select a desired K such that the difference between consecutive iterates is sufficiently small while the iteration is not too long.

We illustrate our proposed CondiQuant in Fig. 3. With the proposed CondiQuant, we convert the weight metric from the quantization-unfriendly position into a friendly position. Besides, both gradient descent and proximal descent steps are calculation efficient and the cost is inexpensive. Moreover, we introduce no additional module. Therefore, no computation and storage overhead is caused during the inference stage and we obtain the theoretic optimal compression and speedup ratio in post-training quantization. After the conversion, we leverage the previous work’s scheme (Liu et al., 2024) to perform quantization and calibration, and the details are described in the supplementary materials.

4. Experiments

To exhibit the outstanding performance of our proposed CondiQuant, comparisons with SOTA methods are also provided in both quantitative and qualitative forms. Extensive ablation studies are conducted to show the robustness and effectiveness of our elaborate designs.

4.1. Experimental Settings

Data and Evaluation. We employ DF2K (Timofte et al., 2017; Lim et al., 2017) as the calibration set, which combines DIV2K (Timofte et al., 2017) and Flickr2K (Lim et al., 2017). During calibration, we only use low-resolution ones and the FP model. Thereafter, CondiQuant is tested on five commonly used benchmarks in the SR field, including Set5 (Bevilacqua et al., 2012), Set14 (Zeyde et al., 2010), B100 (Martin et al., 2001), Urban100 (Huang et al., 2015), and Manga109 (Matsui et al., 2017). The evaluation metrics include PSNR and SSIM (Wang et al., 2004), which are calculated on the Y channel of the YCbCr space.

Implementation Details. The backbone of CondiQuant is SwinIR-light (Liang et al., 2021), a small and efficient ViT for image restoration. The scale factors include $\times 2$, $\times 3$, and $\times 4$, and bit-width includes 2, 3, and 4 bits. During condition number optimization, the calibration set size is 100 images, randomly selected from DF2K and cropped to $3 \times 64 \times 64$. We set step size $\eta = 10^{-2}$, regularization coefficient $\lambda = 0.003$, $\mu = 1$, and max iteration $K = 50$. The target value t is the mean value of singular values. Our code is written with PyTorch (Paszke et al., 2019) and runs on an NVIDIA A800-80GB GPU.

4.2. Ablation Study

In this section, we conduct four ablation studies on Set5 ($\times 2$) with 2 bits to evaluate our design.

Learning Rate η . In CondiQuant, η decides the step size of the gradient descent step. We evaluate η from 10^{-4} to 10^{-1} . As shown in Tab. 2a, the impact of η is minor in the evaluation range. We attribute this minor impact to the rank deficiency across the network and the design of proximal descent. Rank deficiency allows multiple candidates for W while the SVD reconstruction also guarantees the output remains the same. Therefore, the difference between $W_{k+\frac{1}{2}}$ and W_k is small enough. However, the gradient step is not unnecessary as it ensures the output won’t change much.

Selection of t . As discussed in Sec. 3.3, there are several ways to select the target value t in $\mathcal{G}(W)$, including the mean value and median value of all singular values and $(\sigma_{min} + \sigma_{max})/2$. We try these variants and the result is shown in Tab. 2b. The stable performance with different variants indicates that, as long as the singular values are compacted, the performance could therefore be improved. Therefore, t is assigned with the mean singular values.

Regularization Coefficient λ . As shown in Tab. 2c, we test different λ from 0.001 to 0.01. The optimal results could be obtained when λ is set to 0.004. Other settings also provide good results, showing the robustness of CondiQuant.

4.3. Comparison with State-of-the-Art Methods

We adopt two kinds of SOTA PTQ methods for comparison. The first kind is PTQ methods specifically for SR, including DBDC+Pac (Tu et al., 2023), 2DQuant (Liu et al., 2024). The second kind includes PTQ4ViT (Yuan et al., 2022), RepQ (Li et al., 2023), and NoisyQuant (Liu et al., 2023), which are designed for general vision transformers.

Quantitative Results. Table 3 shows the comprehensive results comparing with edge PTQ methods with 2 $\sim$ 4 bits and 2 $\sim$ 4 scale factors on five common SR benchmarks. With two universally accepted metrics, our proposed CondiQuant outperforms all other methods on all benchmarks regardless of bit-width and scale factor. Specifically, CondiQuant has a huge improvement of 1.11 dB and 1.60 dB on $\times 4$, 2-bit on Set5 and Manga109 datasets, respectively. Besides, the average difference between $\times 2$, 4 bits CondiQuant, and the full-precision model is 0.38 dB. This means the quantization degradation when compressing to only 4 bits is minimal and small enough for real-world deployment on edge devices.

Table 3. Quantitative comparison with SOTA methods. The first and second highest methods are marked with **red** and **blue** respectively. Our proposed CondiQuant remarkably outperforms **all** other methods in **all** settings on **all** benchmarks.

Method	Bit	Set5 ($\times 2$)		Set14 ($\times 2$)		B100 ($\times 2$)		Urban100 ($\times 2$)		Manga109 ($\times 2$)	
		PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
SwinIR-light (Liang et al., 2021)	32	38.15	0.9611	33.86	0.9206	32.31	0.9012	32.76	0.9340	39.11	0.9781
Bicubic	32	32.25	0.9118	29.25	0.8406	28.68	0.8104	25.96	0.8088	29.17	0.9128
DBDC+Pac (Tu et al., 2023)	4	37.18	0.9550	32.86	0.9106	31.56	0.8908	30.66	0.9110	36.76	0.9692
PTQ4ViT (Yuan et al., 2022)	4	37.43	0.9571	33.19	0.9139	31.84	0.8950	31.54	0.9212	37.59	0.9735
RepQ (Li et al., 2023)	4	37.89	0.9600	33.47	0.9174	32.08	0.8975	31.98	0.9269	38.37	0.9763
NoisyQuant (Liu et al., 2023)	4	37.50	0.9570	33.06	0.9101	31.73	0.8936	31.31	0.9181	37.47	0.9723
2DQuant (Liu et al., 2024)	4	37.87	0.9594	33.41	0.9161	32.02	0.8971	31.84	0.9251	38.31	0.9761
CondiQuant (ours)	4	38.03	0.9605	33.50	0.9180	32.16	0.8993	32.03	0.9282	38.57	0.9769
DBDC+Pac (Tu et al., 2023)	3	35.07	0.9350	31.52	0.8873	30.47	0.8665	28.44	0.8709	34.01	0.9487
PTQ4ViT (Yuan et al., 2022)	3	36.49	0.9510	32.49	0.9045	31.27	0.8854	30.16	0.9027	36.41	0.9673
RepQ (Li et al., 2023)	3	35.06	0.9325	31.29	0.8719	30.04	0.8512	29.17	0.8726	34.89	0.9518
NoisyQuant (Liu et al., 2023)	3	35.32	0.9334	31.88	0.8911	30.73	0.8710	29.28	0.8835	35.30	0.9537
2DQuant (Liu et al., 2024)	3	37.32	0.9567	32.85	0.9106	31.60	0.8911	30.45	0.9086	37.24	0.9722
CondiQuant (ours)	3	37.77	0.9594	33.21	0.9151	31.94	0.8966	31.18	0.9197	38.01	0.9755
DBDC+Pac (Tu et al., 2023)	2	34.55	0.9386	31.12	0.8912	30.27	0.8706	27.63	0.8649	32.15	0.9467
PTQ4ViT (Yuan et al., 2022)	2	33.25	0.8923	30.22	0.8402	29.21	0.8066	27.31	0.8111	32.75	0.9093
RepQ (Li et al., 2023)	2	31.65	0.8327	29.19	0.7789	28.27	0.7414	26.56	0.7455	30.46	0.8268
NoisyQuant (Liu et al., 2023)	2	30.13	0.7620	28.80	0.7556	28.26	0.7421	26.68	0.7627	30.40	0.8204
2DQuant (Liu et al., 2024)	2	36.00	0.9497	31.98	0.9012	30.91	0.8810	28.62	0.8819	34.40	0.9602
CondiQuant (ours)	2	37.15	0.9567	32.74	0.9103	31.55	0.8912	29.96	0.9047	36.63	0.9713
Method	Bit	Set5 ($\times 3$)		Set14 ($\times 3$)		B100 ($\times 3$)		Urban100 ($\times 3$)		Manga109 ($\times 3$)	
		PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
SwinIR-light (Liang et al., 2021)	32	34.63	0.9290	30.54	0.8464	29.20	0.8082	28.66	0.8624	33.99	0.9478
Bicubic	32	29.54	0.8516	27.04	0.7551	26.78	0.7187	24.00	0.7144	26.16	0.8384
DBDC+Pac (Tu et al., 2023)	4	33.42	0.9143	29.69	0.8261	28.51	0.7869	27.05	0.8217	31.89	0.9274
PTQ4ViT (Yuan et al., 2022)	4	33.77	0.9201	29.75	0.8272	28.62	0.7942	27.43	0.8361	32.50	0.9360
RepQ (Li et al., 2023)	4	34.08	0.9232	30.04	0.8345	28.88	0.8013	27.87	0.8462	32.98	0.9401
NoisyQuant (Liu et al., 2023)	4	33.13	0.9122	29.06	0.8093	27.93	0.7754	26.66	0.8143	31.94	0.9293
2DQuant (Liu et al., 2024)	4	34.06	0.9231	30.12	0.8374	28.89	0.7988	27.69	0.8405	32.88	0.9389
CondiQuant (ours)	4	34.32	0.9260	30.29	0.8417	29.05	0.8039	28.05	0.8506	33.23	0.9431
DBDC+Pac (Tu et al., 2023)	3	30.91	0.8445	28.02	0.7538	26.99	0.6937	25.10	0.7122	28.84	0.8403
PTQ4ViT (Yuan et al., 2022)	3	32.75	0.9028	29.14	0.8113	28.06	0.7712	26.43	0.8014	31.20	0.9131
RepQ (Li et al., 2023)	3	31.04	0.8548	28.04	0.7572	26.83	0.7019	25.56	0.7493	30.16	0.8904
NoisyQuant (Liu et al., 2023)	3	30.78	0.8511	27.94	0.7624	26.98	0.7153	25.43	0.7481	29.64	0.8792
2DQuant (Liu et al., 2024)	3	33.24	0.9135	29.56	0.8255	28.50	0.7873	26.65	0.8116	31.46	0.9235
CondiQuant (ours)	3	33.92	0.9224	30.02	0.8367	28.84	0.7986	27.37	0.8356	32.48	0.9367
DBDC+Pac (Tu et al., 2023)	2	29.96	0.8254	27.53	0.7507	27.05	0.7136	24.57	0.7117	27.23	0.8213
PTQ4ViT (Yuan et al., 2022)	2	29.96	0.7901	27.36	0.7030	26.74	0.6590	24.56	0.6460	27.37	0.7390
RepQ (Li et al., 2023)	2	27.32	0.6478	25.63	0.5918	25.44	0.5652	23.42	0.5582	24.51	0.5721
NoisyQuant (Liu et al., 2023)	2	27.53	0.6641	25.77	0.5952	25.37	0.5613	23.59	0.5739	26.03	0.6632
2DQuant (Liu et al., 2024)	2	31.62	0.8887	28.54	0.8038	27.85	0.7679	25.30	0.7685	28.46	0.8814
CondiQuant (ours)	2	33.00	0.9130	29.44	0.8253	28.45	0.7882	26.36	0.8080	30.88	0.9203
Method	Bit	Set5 ($\times 4$)		Set14 ($\times 4$)		B100 ($\times 4$)		Urban100 ($\times 4$)		Manga109 ($\times 4$)	
		PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
SwinIR-light (Liang et al., 2021)	32	32.45	0.8976	28.77	0.7858	27.69	0.7406	26.48	0.7980	30.92	0.9150
Bicubic	32	27.56	0.7896	25.51	0.6820	25.54	0.6466	22.68	0.6352	24.19	0.7670
DBDC+Pac (Tu et al., 2023)	4	30.74	0.8609	27.66	0.7526	26.97	0.7104	24.94	0.7369	28.52	0.8697
PTQ4ViT (Yuan et al., 2022)	4	31.49	0.8831	28.04	0.7680	27.20	0.7240	25.53	0.7660	29.52	0.8940
RepQ (Li et al., 2023)	4	31.77	0.8880	28.32	0.7750	27.40	0.7310	25.83	0.7780	29.88	0.9010
NoisyQuant (Liu et al., 2023)	4	31.09	0.8751	27.75	0.7591	26.91	0.7151	25.07	0.7500	28.96	0.8820
2DQuant (Liu et al., 2024)	4	31.77	0.8867	28.30	0.7733	27.37	0.7278	25.71	0.7712	29.71	0.8972
CondiQuant (ours)	4	32.09	0.8923	28.50	0.7792	27.52	0.7345	25.97	0.7831	30.16	0.9054
DBDC+Pac (Tu et al., 2023)	3	27.91	0.7250	25.86	0.6451	25.65	0.6239	23.45	0.6249	26.03	0.7321
PTQ4ViT (Yuan et al., 2022)	3	29.77	0.8337	27.00	0.7248	26.21	0.6735	24.22	0.6983	27.94	0.8479
RepQ (Li et al., 2023)	3	27.52	0.7419	24.84	0.5996	23.99	0.5351	22.42	0.5739	26.58	0.7838
NoisyQuant (Liu et al., 2023)	3	28.90	0.7972	26.50	0.6970	26.16	0.6628	23.86	0.6667	27.17	0.8116
2DQuant (Liu et al., 2024)	3	30.90	0.8704	27.75	0.7571	26.99	0.7126	24.85	0.7355	28.21	0.8683
CondiQuant (ours)	3	31.62	0.8855	28.20	0.7715	27.31	0.7269	25.39	0.7624	29.29	0.8915
DBDC+Pac (Tu et al., 2023)	2	25.01	0.5554	23.82	0.4995	23.64	0.4544	21.84	0.4631	23.63	0.5854
PTQ4ViT (Yuan et al., 2022)	2	27.23	0.6702	25.38	0.5914	25.15	0.5621	22.94	0.5587	24.66	0.6132
RepQ (Li et al., 2023)	2	25.55	0.5834	23.54	0.4751	23.30	0.4298	21.62	0.4493	23.60	0.5561
NoisyQuant (Liu et al., 2023)	2	25.94	0.5862	24.33	0.5067	24.16	0.4718	22.32	0.4841	23.82	0.5403
2DQuant (Liu et al., 2024)	2	29.53	0.8372	26.86	0.7322	26.46	0.6927	23.84	0.6912	26.07	0.8163
CondiQuant (ours)	2	30.64	0.8671	27.59	0.7567	26.93	0.7136	24.54	0.7282	27.67	0.8613

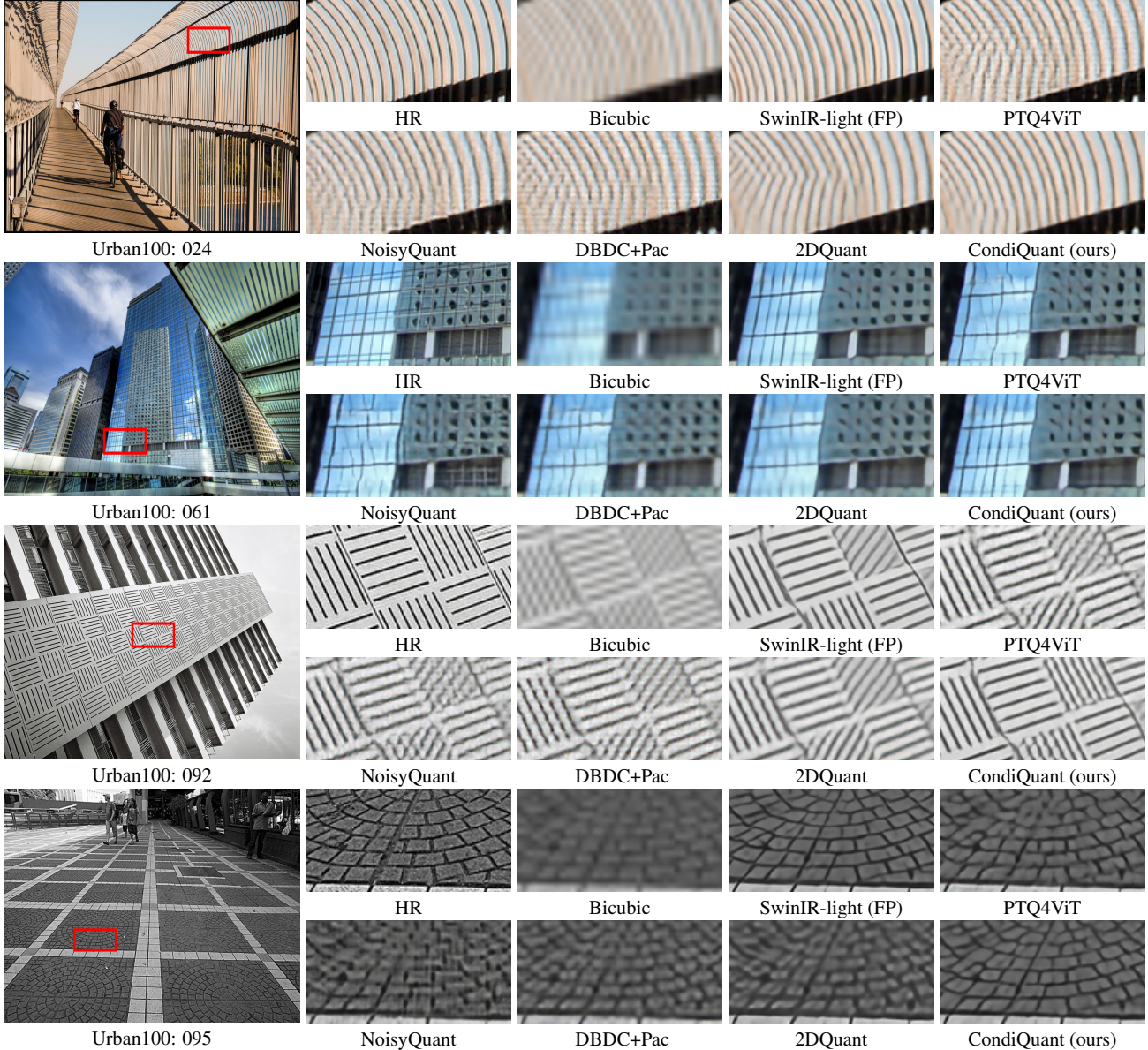


Figure 4. Visual comparison for image SR. We compare our proposed CondiQuant with current competitive quantization methods and the full-precision (FP) model. The visual results illustrate that CondiQuant gains sharper edges and reasonable textures.

With condition number optimization, the model’s sensitivity to the quantization error is reduced. Thereafter, the weight matrix is efficiently converted into a more quantization-friendly representation. Furthermore, the quantized model enjoys significant improvement during the following calibration and distillation stage. Besides, the oscillation during distillation is reduced, and detailed supporting information is in the supplemental material.

Visual Results. Regularly occurring dense and fine textures are **typical challenging scenarios** for image super-resolution. As shown in Fig. 4, we choose the most challenging cases to validate the visual effect of CondiQuant with $\times 4$. The competing methods are struggling to recover erratic textures and often generate artifacts. On the contrary,

with our proposed CondiQuant, the quantized model could reconstruct HR with rich details and reasonable structures. In 024 and 061, the direction of the texture is distorted, and jagged edges are generated with Bicubic. Given the misleading input, other methods can not provide robust restoration like the FP model. However, our proposed CondiQuant can still guarantee fidelity to the greatest extent.

Even more, in 092, the FP model can not provide results with accurate direction on the wall, while CondiQuant can correct the error and provide sharp contents. This is because the FP model suffers from over-fitting, and CondiQuant could ease this phenomenon. Besides, in 095, the competing methods recover bricks with vague texture while CondiQuant remains aligned with the FP model. In conclusion, compared to other

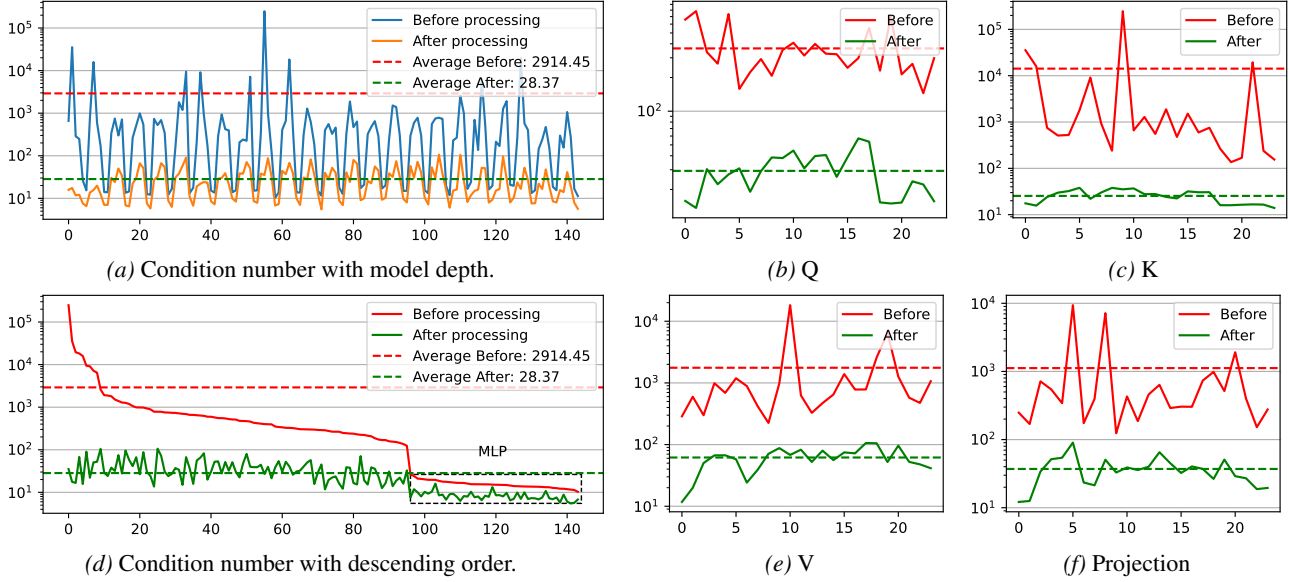


Figure 5. Condition number before and after CondiQuant on $\times 2$ model. Before CondiQuant, the distribution of condition numbers is extremely high, with an average of nearly 3k. After CondiQuant, the average value of condition numbers is significantly reduced to 28.37.

Table 4. Performance on Manga109 ($\times 4$) with 4 bits.

	SwinIR		MambaIRv2		EDSR	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
FP	30.92	0.9150	31.24	0.9182	30.35	0.9070
2DQuant	29.71	0.8972	28.30	0.8540	28.95	0.8143
CondiQuant	30.16	0.9054	29.75	0.9007	29.24	0.8575

methods, CondiQuant achieves the minimum information loss after low bit quantization and restores images with high-fidelity, rich details, and sharper edges.

4.4. Condition Number Analysis

The excellent results are closely related to the decrease in condition number. Therefore, we visualize the condition number shifts in Fig. 5 on the SR model with the scale factor of 2. Figures 5a and 5d show the condition number of all layers (including Q, K, V, projection, FC1, and FC2) with different sequences. Fig. 5a is about model depth, and Fig. 5d sorts the condition number before CondiQuant with descending order. The other four figures show different layers with model depth as the X-axis.

Overall, the average condition number is hugely reduced from nearly 3k to merely 28.37. This evident decrease shows the effectiveness of the proximal descent step in CondiQuant. To be specific, a regular pattern is observed with model depth. The condition numbers of the FC1 and FC2 in the MLP layer are distinctly smaller than those of other layers, and their influence is also minor. Hence, considering efficiency, we do not perform CondiQuant on FC1 and FC2. The condition number of the K matrix in self-attention is usually extremely large. This is also consistent with the claim that the quantization degradation of ViT is mainly attributed to self-attention. Moreover, the distribution of the condition number after CondiQuant is greater in the middle

and smaller at both ends. This indicates that the beginning and end modules are more important and sensitive in image restoration. To conclude, CondiQuant significantly reduces the condition number with high efficiency.

4.5. Architecture Generalization

To validate the generalization on different architectures, the performance of CondiQuant on three different architectures is tested on Manga109 with 4bit at a scale factor of 4. Three representative architectures are selected. SwinIR (Liang et al., 2021) is for vision transformer, MambaIRv2 (Guo et al., 2025) is for Mamba, and EDSR (Lim et al., 2017) is for CNN. The consistently higher metrics represent the generalization ability across different architectures.

5. Conclusion

In this paper, we propose CondiQuant, a condition number based PTQ method for image super-resolution. We analyze that the degradation of quantization is attributed to the quantization of activation and leverage condition number to minimize the impact of the activation quantization error. Thereafter, we formulate the optimization problem to minimize the condition number while maintaining the output as it is. To optimize, we design the gradient descent step to keep the output still and the proximal descent step to reduce the condition number. Both steps are calculation-efficient and the entire iteration process takes only 19.0 seconds. As there’s no additional module, we reach the theoretically optimal compression and speedup ratio. Specifically, when quantized to 2 bits, the compression ratio is $3.60\times$ and the speedup ratio is $5.08\times$. The comparison experiments demonstrate the excellent performance of CondiQuant while the ablation studies present its robustness.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. A conclusion of the motivation

It is a necessary step to perform model quantization on the SR networks before deployment. However, the quantization process leads to performance degradation. The degradation is mainly from the activation, as the features change according to different input images. However, the quantization error of activation is difficult to minimize, as it is strongly correlated with the input.

Therefore, we decided to minimize the **impact** of the quantization error of activation (not the error itself), which is represented by the condition number of the weight matrix. When the condition number is lower, the impact is smaller, even when the activation quantization error remains unchanged. As most of the activation is rank-deficient, we can shift the weight matrix without introducing additional error. Therefore, we propose CondiQuant to reduce the condition number of the weight matrix while keeping the output unchanged.

B. Derivation of proximal gradient descent

In convex optimization problems, for differentiable objective functions, we can iteratively find the optimal solutions simply using gradient descent. For non-differentiable objective functions, we can still iterate toward the optimal solution by introducing subgradients. However, subgradient methods are much slower compared to gradient descent ones. To address this, for functions that are non-differentiable in their entirety but can be separated into differentiable and non-differentiable components, we can use a faster algorithm known as the proximal gradient descent method. Generally, consider an objective function that can be decomposed into two functions of the following forms:

$$f(x) = g(x) + h(x), \quad (17)$$

where $g(x)$ is a convex and differentiable function, while $h(x)$ is also convex but non-differentiable function. Using proximal gradient descent, a convergence rate of $O(1/\epsilon)$ can be achieved, where $\epsilon = f(x^k) - f(x^*)$ represents the deviation between the current iteration result and the optimal solution. Moreover, by accelerating the proximal gradient, a convergence rate of $O(1/\sqrt{\epsilon})$ can be attained.

General form. For the differentiable part $g(x)$, assuming the starting point is at x_0 , the second-order Taylor expansion can be expressed as:

$$g(x) \approx g(x_0) + \nabla g(x_0)^T (x - x_0) + \frac{1}{2} \nabla^2 g(x_0) (x - x_0)^2. \quad (18)$$

Assuming $g(x)$ satisfies L -Lipschitz continuity, where the upper bound of the Hessian matrix is L . By setting the step size $\eta = \frac{1}{L}$, the second-order term can be simplified to $\frac{1}{2\eta}$. The problem can be written as:

$$\begin{aligned} \min_x \frac{1}{2\eta} [(x - x_0)^2 + 2\eta \nabla g(x_0)(x - x_0) + (\eta \nabla g(x_0))^2] \\ + h(x) \\ = \min_x \frac{1}{2\eta} [x - x_0 + t \nabla g(x_0)]^2 + h(x) \\ = \min_x \frac{1}{2\eta} \|x - (x_0 - t \nabla g(x_0))\|_2^2 + h(x) \\ = \min_x \frac{1}{2\eta} \|x - z\|_2^2 + h(x), \end{aligned}$$

where $z = x_0 - t \nabla g(x_0)$ is the form of gradient descent for $g(x)$. Therefore, its solution is:

$$x = \arg \min_x \frac{1}{2} \|x - z\|_2^2 + \eta g(x), \quad (19)$$

Eq. 19 is also known as the proximal operator, denoted as:

$$\text{prox}_{\eta, g}(z) = \arg \min_x \frac{1}{2} \|x - z\|_2^2 + \eta \cdot h(x). \quad (20)$$

Using Proximal Functions, we can define an iterative process for optimization, called the proximal descent method. First, choose an initial point x_0 and then iterate:

$$x_i = \text{prox}_{h, \eta}(x_{i-1} - \eta_i \nabla g(x_{i-1})), \quad i = 1, 2, 3, \dots \quad (21)$$

There are some advantages of using this method:

1. For many h functions, the proximal projection $\text{prox}_{h, \eta}$ has an analytical solution.
2. The $\text{prox}_{h, \eta}$ depends only on h , so it can be used for different functions g .
3. g can be any convex function, as long as we can compute its gradient.

1 CondiQuant pipeline

Input: Pretrained weight matrix W ; Activation X from calibration set; Learning rate or step size η ; Penalty parameter λ ; Maximum iterations K .

Output: Quantization friendly weight matrix W' .

```

1  $U, \Sigma, V \leftarrow \text{SVD}(W)$ 
2  $Y \leftarrow XW$ 
3 for  $k = 1$  to  $K$  do
4    $W_{k+\frac{1}{2}} \leftarrow W_k - \eta \nabla_W \frac{1}{2} \|Y - XW\|_F^2 \Big|_{W=W_k}$ 
5    $U_k, \Sigma_k, V_k \leftarrow \text{SVD}(W_k)$ 
6    $t \leftarrow \text{mean}(W_k)$ 
7    $\Sigma_k^* \leftarrow (\Sigma_k + 2\lambda\mu t)/(1 + 2\lambda\mu)$ 
8    $W_{k+1} \leftarrow U_k \Sigma_k^* V_k$ 
9    $k \leftarrow k + 1$ 
10 end for
11  $W' \leftarrow W_K$ 
12 return  $W'$ 
    
```

Accelerated proximal gradient method. Choose an initial point $x_{-1} = x_0$, and then repeat the following steps:

1. Use the result from the previous step as the momentum and calculate z :

$$z = x_{k-1} + \frac{k-2}{k+1} (x_{k-1} - x_{k-2}), \quad (22)$$

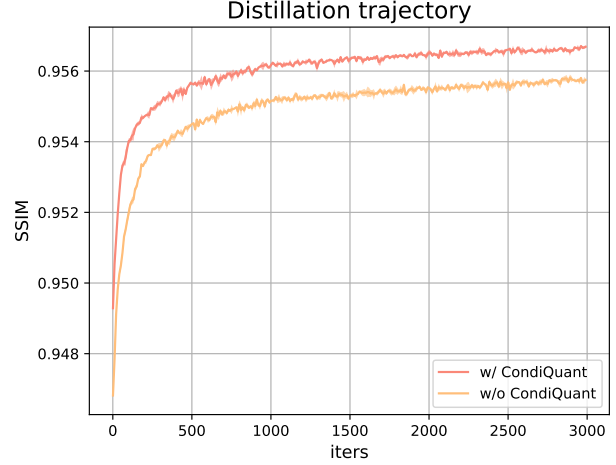
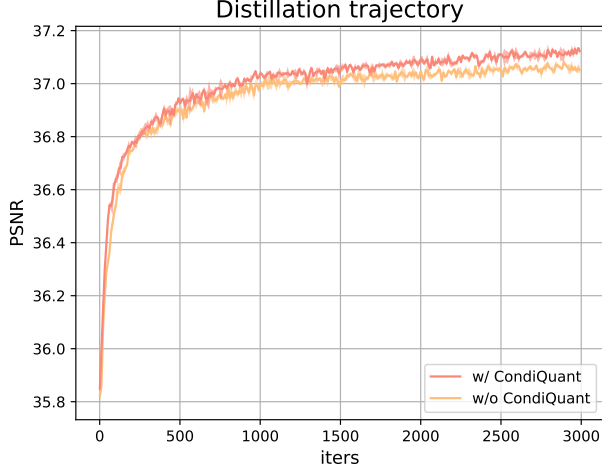
2. Update x using z :

$$x_k = \text{prox}_{\eta_k}(z - \eta_k \nabla g(z)), \quad (23)$$

The first step of $k = 1$ is the standard proximal gradient update. Afterward, we use z to replace x_{k-1} , transferring momentum from previous iterations to update the next gradient direction, making it align more closely with the current gradient.

C. Calibration details

Implementation details. We adopted a per-channel granularity and quantize all the linear layers within the transformer blocks. Although some layers are full-rank, we still perform CondiQuant, given their limited impact. We made some modifications to the DOBI (Distribution-oriented bound initialization) algorithm of 2DQuant (Liu et al., 2024) simultaneously. Specifically, for the long-tail distribution activations, we narrow the search range to 10 evenly spaced numbers between the 90th percentile and the maximum value. For weights and activations that follow an approximately symmetric distribution, we will search both the upper and lower bounds. The upper bound search range is from the 90th percentile to the maximum value, while the lower bound range is from the minimum value to the 10th percentile. We will also search 10 values within each range.



(a) Comparison of distillation trajectories of PSNR on Set5.

(b) Comparison of distillation trajectories of SSIM on Set5.

Figure 6. Condition number before and after CondiQuant on $\times 2$ model. Before CondiQuant, the distribution of condition numbers is extremely high, with an average of nearly 3k. After CondiQuant, the average value of condition numbers is significantly reduced to 28.37.

Visualization of distillation. We provide more visual comparison results in knowledge distillation to demonstrate the role of CondiQuant. It is performed based on 2-bit quantization of SwinIR-light $\times 4$. We changed different random seeds for multiple distillations and obtained the distillation trajectory with a 95% confidence interval as shown in Fig. 6a and Fig. 6b.

D. The ignored second-order term

In the main paper, we decompose the quantization error in the following form

$$\begin{aligned}\hat{Y} &:= \hat{X}\hat{W} = (X + \delta X)(W + \delta W) \\ &= XW + X\delta W + \delta XW + \delta X\delta W.\end{aligned}\quad (24)$$

The second-order $\delta X\delta W$ is directly ignored because the impact is minimal even when the bit-width is low. With 2

bits, for example, we have $|\delta W|_F \leq \frac{1}{4}|W|_F$. Such that we informally have

$$|\delta X\delta W|_F^2 \approx |\delta X|_F^2 |\delta W|_F^2 \ll |\delta X|_F^2 |W|_F^2 \approx |\delta XW|_F^2. \quad (25)$$

With Eq (25), the second-order term is much smaller than the first-order term, meaning that the second-order term can be safely ignored.

E. More visual results

We provide more visual comparison results in Figures 7 and 8. All results demonstrate the outstanding performance of our proposed CondiQuant.

2 Modified DOBI pipeline

Input: Data to be quantized v , bit b , number of search point K .

Output: Clip bound l, u

```

1  $l \leftarrow \min(v)$ 
2  $u \leftarrow \max(v)$ 
3  $min\_mse \leftarrow +\infty$ 
4 if  $v$  is symmetrical then
5    $v_{10} \leftarrow$  10th percentile of  $v$ 
6    $\Delta l \leftarrow (v_{10} - \min(v))/K$ 
7 else
8    $\Delta l \leftarrow 0$ 
9 end if
10  $v_{90} \leftarrow$  90th percentile of  $v$ 
11  $\Delta u \leftarrow (\max(v) - v_{90})/K$ 
12 for  $i = 1$  to  $K$  do
13    $l_i \leftarrow l + i \times \Delta l$ 
14    $u_i \leftarrow u - i \times \Delta u$ 
15   get  $v_q$  based on quantization formula.
16    $mse \leftarrow \|v - v_q\|_2$ 
17   if  $mse \leq min\_mse$  then
18      $min\_mse \leftarrow mse$ 
19      $l\_best \leftarrow l_i, u\_best \leftarrow u_i$ 
20   end if
21 end for
22 return  $l\_best, u\_best$ 
```

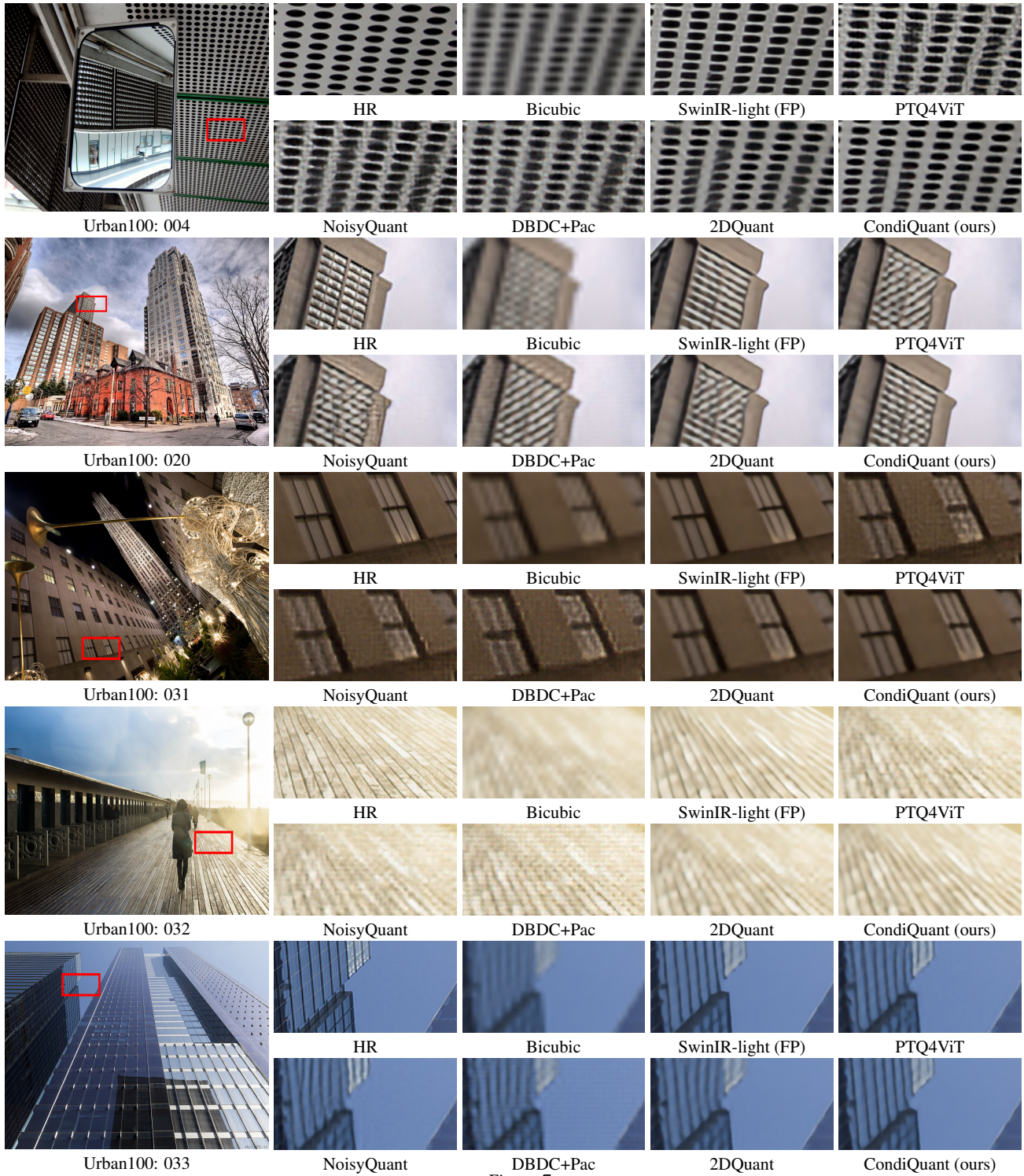


Figure 7

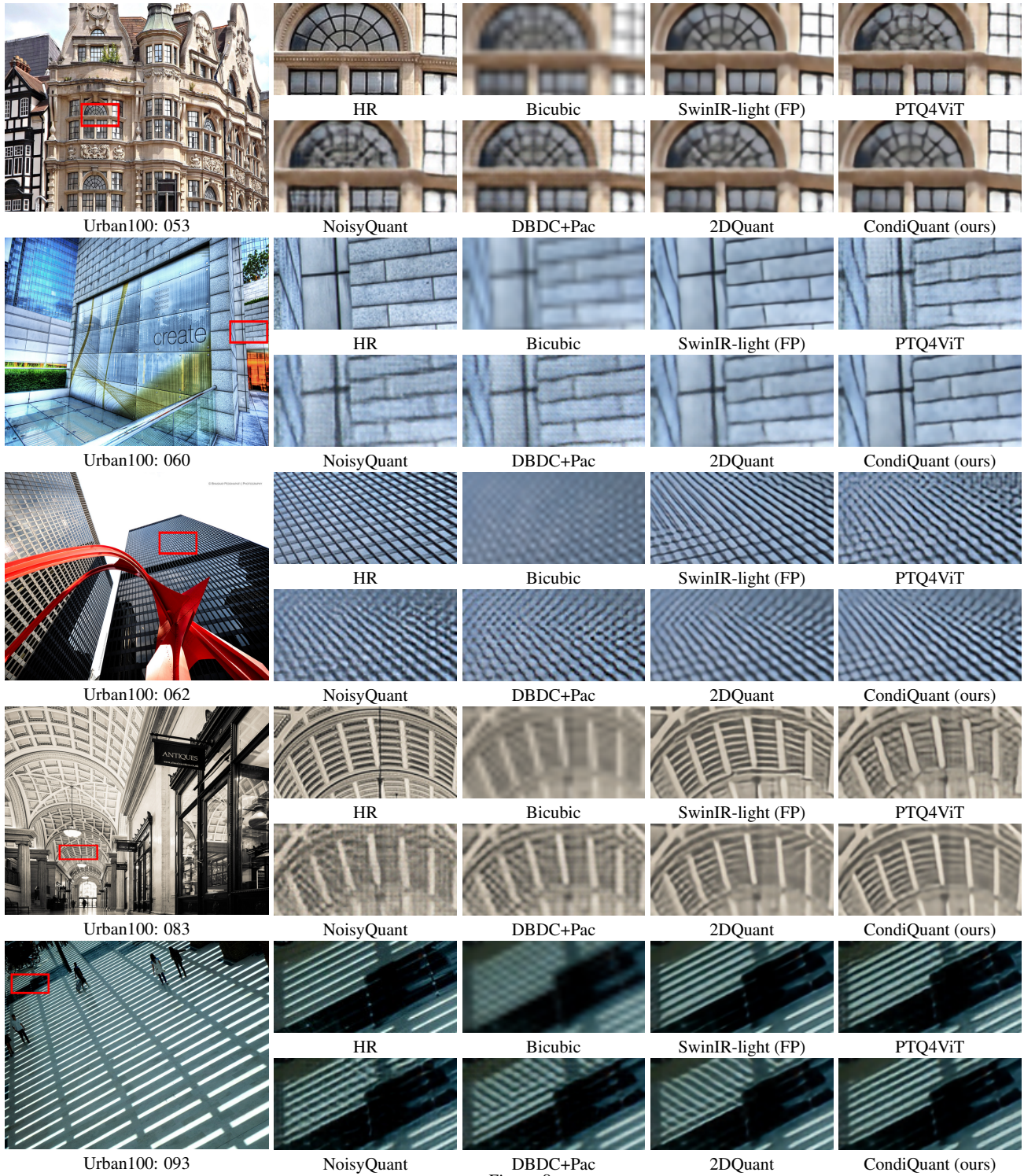


Figure 8